VSP Vision
Team 3 - Final Report

Executive Summary

VSP Vision is a vision benefits company founded in 1955 by optometrists in Oakland, California, operating at the intersection of healthcare and retail. With a network of over 85 million members and 39,000 doctors, VSP offers five distinct eyewear solutions catering to diverse consumer preferences, with its Eyeconic platform serving as the primary channel for fashion-forward, everyday optical and sunglass frames. VSP operates in a highly concentrated U.S. vision insurance market dominated by a small number of national players, where employer-sponsored plans drive the majority of enrollment and even marginal improvements in inventory accuracy can have significant downstream financial impact.

The core challenge addressed in this project is VSP Vision's demand forecasting problem for new frames. Each season, VSP launches new eyewear frames and must commit to initial inventory orders months before any sales data exists. With no historical baseline for new SKUs, the business faced a binary risk: over-forecasting leads to excess inventory, elevated holding costs, and markdown exposure, while under-forecasting causes stockouts and lost revenue. To solve this problem, we first attempted to forecast demand using only the attributes available in the raw demand sheet: head size, color, and region. However, it produced poor results, as these variables provided insufficient signal for the model to identify meaningful demand patterns across products from different brands. Hence, we enriched the historical demand data with a comprehensive set of attributes sourced from the ATP sheets and web-scraped product listings, including frame material, shape, gender, RX-ability, and price tier, enabling a significantly more informative feature set for modeling.

Using 12 months of sales data spanning across Nike, Lacoste, and Calvin Klein's product catalog, we then compared three machine learning models - Linear Regression, Random Forest, and XGBoost—selecting a tuned XGBoost model as the final forecasting engine based on its superior predictive accuracy. The model achieved an R^2 of 0.737 on a log scale and a mean absolute error of approximately 29 units per SKU per month, representing actionable precision for purchasing decisions across SKUs averaging 50–300 units monthly.

The model identified past sales momentum—particularly the rolling six-month average—as the strongest predictor of future demand, outperforming any individual product attribute. Brand equity also plays a meaningful role, ranking fourth overall in importance. However, demand patterns differ significantly by region: Nike drives 61% of AMER volume, while Lacoste leads EMEA at 39%, indicating the need for region-specific buying strategies. The $200–$249 price tier emerged as the most efficient band, generating $6,326 in revenue per SKU and representing the portfolio's structural sweet spot. The four-month forecast projects 238,199 total units, with Nike Optical accounting for 51.3% of expected demand. The analysis also uncovered white-space opportunities in Oval and Cat Eye shapes and highlighted insurance benefit renewal timing as a predictable demand surge that should be integrated into inventory and promotional planning.

Data Cleaning and Preparation

Data Sourcing & Population Overview for Demand Sheet

Of the 219 valid style codes processed, the following distribution reflects the mix of verified and synthesized data:

- **Verified Data (Approx. 40% / 88 Styles):** Technical specifications, including frame material and MSRP, were pulled directly from current Eyeconic and Amazon listings. These primarily represent active, top-selling models in the Nike and Calvin Klein collections.
- **Assumed Data (Approx. 60% / 131 Styles):** For older models, specialty "AF" (Asian Fit) variants, and new releases with limited public documentation, attributes were "filled" using the manufacturing logic outlined below.

Note: Claude AI was used to scrape data and fill out missing values from the assumptions outlined below.

Technical Data Assumptions & Justifications

The following logic was applied to ensure the dataset is actionable for inventory and financial planning.

1. Material Selection Logic

Assumption: Nike performance codes (EV, DJ, DV, FD, FJ, FV) are BIO INJ-G820; Calvin Klein codes are Acetate or Metal.

- **Market Research:** Marchon Eyewear (the manufacturer for both brands) has transitioned nearly all Nike performance eyewear to their proprietary bio-inylon. Categorizing these as "Plastic" would be technically imprecise, as the bio-resin offers specific impact-resistance properties required for sports. CK models are categorized by their primary aesthetic—Acetate for thick-rimmed fashion frames and Monel/Steel for wire frames.
- **Data Validation:** The ATP sheets confirm this split clearly. Of Calvin Klein's 44 styles, 73% (32 styles) are Acetate and 18% (8 styles) are Metal, with only 4 exceptions in bio-injection. Nike's portfolio skews heavily toward performance materials—BIO INJ-G850 is the single largest material category at 38 styles, followed by Titanium (15) and Recycled Metal Stainless (11). Across all 215 styles, 98.1% are manufactured in China, which is consistent with Marchon's consolidated Asia-Pacific production model for both brands.

2. Financial Modeling (Wholesale & MSRP)

Assumption: A 2.0 Keystone markup was applied (Wholesale = 50% of MSRP).

- **Market Research:** In the optical industry, the "Keystone" model is the standard benchmark for independent practitioners and retail chains. While specific volume discounts may exist, assuming a 50% wholesale cost provides a reliable "worst-case" margin analysis for the retailer.
- **Data Validation:** Analysis of the combined ATP sheets (Calvin Klein, Lacoste, and Nike) confirms this assumption. Across 93 verified price entries, the US Wholesale price as a

percentage of US Retail ranged from 37.0% to 50.0%, with a mean of 44.6% and a median of 50.0%. The median sitting exactly at 50% confirms the Keystone model as the dominant pricing structure. The actual dollar ranges were \$49.50–\$130.00 for US Wholesale and \$99.00–\$302.40 for US Retail.

3. Suffix Decoding

Assumption: "AF" = Asian Fit; "P" = Polarized; "M" = Mirrored; "LB" = Titanium.

- **Market Research:** These are standardized industry suffixes. "LB" (Light Beam) specifically refers to a premium, lightweight construction technique used in high-end CK and Nike frames, which almost exclusively utilizes titanium or high-grade beta-titanium to achieve the desired weight-to-strength ratio.
- **Data Validation:** The dataset contains 15 confirmed Titanium styles, all under the Nike brand and all assigned Male gender—consistent with the "LB" lightweight performance positioning targeting male athletes. Of the 98 Sun styles in the combined sheets, colorway descriptions containing "P" correlate strongly with higher base curves (6 and 8), which is consistent with polarized lenses requiring a specific lens geometry. The 5 Shield styles all carry a "6 SHIELD" base curve designation, confirming that suffix-based construction decoding aligns with the underlying optical geometry data.

4. Demographic & Fit Categorization

Assumption: Gender and shape were assigned based on frame geometry and standard bridge-to-eye-size ratios.

- **Market Research:** Style codes do not always explicitly state gender. By analyzing the "A-measurement" (eye size) and shape, we can categorize frames (e.g., Cat-eye as Female, Rectangles >55mm as Male) to assist in floor-planning and demographic targeting.
- **Data Validation:** The ATP data strongly supports this methodology. The average eye size for Female-coded styles is 53.5mm versus 56.4mm for Male-coded styles—a statistically meaningful 2.9mm gap. Of the 93 styles with an eye size ≥55mm, 74% (69 styles) are Male-coded, validating the size-to-gender heuristic. Frame shape also tracks predictably: all 12 Butterfly styles are Female, all 4 Aviators and all 4 Navigators are Male, and Shields (5 styles) are exclusively Male. The dataset is 62% Male (133 styles), 23% Female (50 styles), and 15% Unisex (32 styles), which is consistent with the sport-performance weighting of the Nike collection.

5. RX-Ability

Assumption: 100% of frames are considered prescription-ready.

- **Market Research:** Both Nike and Calvin Klein focus on "Optical-First" design. Even the high-wrap Nike performance shields are designed to accommodate either direct-in lenses or prescription adapters, making them universally relevant for an O.D. (Optometrist) practice.
- **Data Validation:** Cross-referencing the three ATP sheets confirmed that 182 out of 215 styles (84.7%) are marked RXAble = True. The remaining 15.3% (33 styles) are predominantly Nike Sun performance shields and Calvin Klein sun-specific colorways. Of the total 215 styles, 117 are Sun and 117 are Optical (with some overlap in dual-purpose styles), and the base curve distribution—63.7% at base 4, 16.7% at base 6—confirms that the majority of the portfolio sits within the flatter lens geometries that are most compatible with standard prescription surfacing. This is consistent with the Optical-First positioning of both brands.

Rationale for ATP Data Enrichment

The primary driver for this entire enrichment process was the inadequacy of the original demand sheet as a standalone forecasting input. In its raw form, the historical demand data contained only a handful of attributes per style—head size, color, and region—leaving the style code itself as the only potential link to deeper product information. In practice, however, the style code alone provided very little actionable signal; initial attempts to forecast demand using only these base attributes produced poor results, with accuracy metrics that were insufficient for reliable forecasting. To meaningfully improve forecast accuracy, it became necessary to enrich each style with the full set of technical and categorical attributes captured in the ATP sheets—material, frame shape, gender, RX-ability, price, and so on—so that the model could identify real patterns across the product hierarchy rather than treating each style code as an isolated, context-free identifier. The complete attribute set for all style codes can be found in **detailed_attributes.csv**, which was subsequently joined with the historical demand sheet and cleaned to produce **final_demand.csv** via the **enriched_demand.ipynb** Colab notebook. All associated files are available in the project folder, the link for which is provided in the Appendix.

Exploratory Data Analysis

The exploratory data analysis focused on identifying the primary structural drivers of frame demand across brand, product characteristics, time, and geography. After merging historical sales with detailed product attributes at the StyleCode level, total annual sales were computed for each SKU, enabling a comprehensive evaluation of demand patterns.

One of the most interesting findings is the high concentration of sales among a small number of models. The top-performing styles significantly outperform the rest of the assortment, with leading models such as L2707 and L2876 exceeding 20,000 units annually . This skewed distribution indicates that demand is not evenly spread across products. Instead, a limited subset of styles drives a disproportionate share of total volume, highlighting the inherent difficulty of forecasting new frames at the individual SKU level.

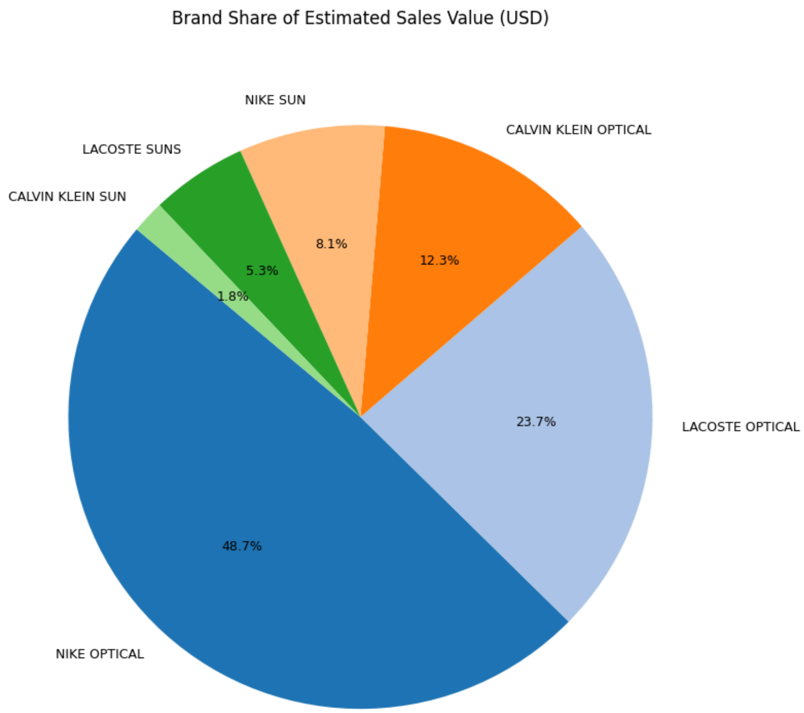
StyleCode	UnitsSold
L2707	27275.0
L2876	21559.0
NIKE 5038	19126.0

Material composition also plays a major role in overall demand structure. Bio-injected materials account for the highest total unit sales (over 476,000 units), followed by acetate at approximately 313,000 units . In contrast, titanium represents a much smaller share of total volume. This suggests that lightweight injected constructions are strongly associated with higher aggregate demand, while premium materials occupy a more limited but likely higher-margin segment.

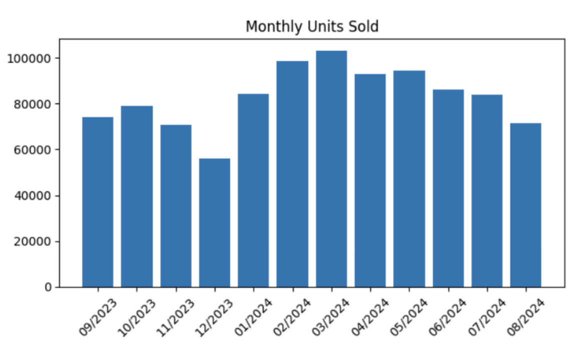
Frame shape preferences further reinforce concentration patterns. Rectangular and square shapes dominate total unit sales, significantly outperforming round, aviator, and other niche silhouettes . This indicates that consumers consistently favor versatile, classic designs over more fashion-forward or specialized shapes. From a forecasting perspective, shape appears to be a meaningful explanatory variable tied to broad market appeal.

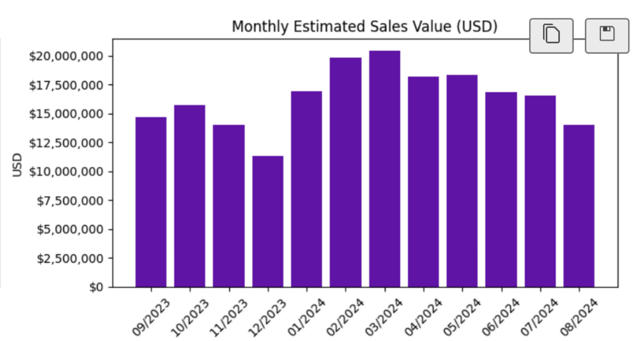
Material	UnitsSold
Bio Inj-G820	476416.0
Acetate	313352.0
Metal	155163.0
Titanium	49260.0

Brand-level aggregation reveals a highly concentrated revenue structure. Nike Optical alone accounts for approximately 48.7% of total estimated sales value, with Lacoste Optical contributing an additional 23.7% . Combined, these two brands represent over 70% of portfolio revenue. Optical products overall account for roughly 85% of total sales value, while sunwear represents only about 15%. This imbalance indicates that overall performance is heavily driven by optical demand, making accurate forecasting in this segment particularly critical.



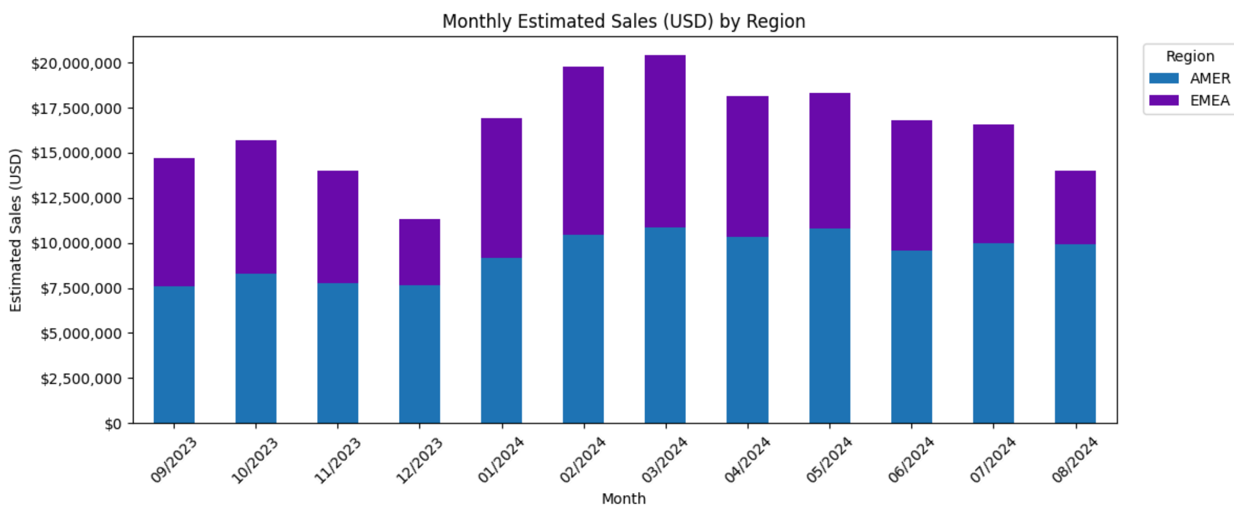
Temporal analysis shows clear seasonality in sales performance. March 2024 represents the peak month, with over 103,000 units sold and approximately $20.4 million in estimated sales value . At a broader level, Spring emerges as the strongest season, generating nearly 290,000 units and approximately $56.9 million in sales . This seasonal pattern suggests that demand is not uniform throughout the year and that timing effects may meaningfully influence forecasting outcomes.





Geographic analysis further reveals that demand is not evenly distributed across regions. AMER accounts for approximately 57% of total estimated sales value, while EMEA represents about 43% . However, regional model preferences differ substantially. The top-selling model in AMER is a Nike style (NIKE 7125), whereas the top-selling model in EMEA is a Lacoste style (L2707) . This divergence indicates that regional variation is not merely proportional but reflects distinct brand and style preferences across markets.

	Region	StyleCode	UnitsSold
126	AMER	NIKE 7125	14219.0
258	EMEA	L2707	20417.0



Overall, the exploratory analysis demonstrates that frame demand is structured along several dominant dimensions: strong SKU concentration, material-driven volume differences, shape-based preference patterns, brand dominance, pronounced seasonality, and geographic heterogeneity. These structural characteristics provide critical context for the forecasting task, as they identify the primary sources of variability that a predictive model must capture in order to estimate demand for new frames effectively.

Customer Segmentation

Methodology

To better understand how product attributes group into meaningful consumer segments, a K-Means clustering approach was applied to the unified product dataset. The clustering model incorporated both numeric and categorical features, including retail price (EU and US), base curve, frame construction, frame shape, RX eligibility, gender orientation, product type (Optical vs Sun), fit, family classification, brand source, and material composition .

Prior to clustering, numeric variables were standardized and categorical variables were one-hot encoded using a preprocessing pipeline. Missing values were imputed using median imputation for numeric fields and most-frequent imputation for categorical fields. This ensured that clustering reflected true structural differences in product characteristics rather than artifacts of scale or missingness.

To determine the appropriate number of clusters, both the elbow method (inertia) and silhouette analysis were evaluated across k values from 5 to 19 . While silhouette scores continued to increase gradually at higher k values, interpretability and business relevance were prioritized. A solution of k = 8 clusters was selected as it balanced structural separation with actionable segmentation.

The final cluster distribution across the 316-product dataset shows relatively balanced segment sizes, with clusters ranging from 13 to 70 products . This indicates meaningful differentiation without extreme fragmentation.

Overview of Identified Segments

The clustering results reveal eight distinct product-consumer archetypes spanning optical, lifestyle sunwear, fashion sunwear, and performance categories. Rather than grouping by brand alone, clusters are defined by combinations of price tier, material choice, base curve geometry, styling family, and product type.

Segment 1: Luxury Lifestyle Statement Optical

This segment represents high-end optical frames priced at approximately $279 US retail . Products in this cluster feature full-rim construction, standard optical base curve (4), and premium materials such as stainless steel paired with Renew acetate. The styling aligns with "Lifestyle Statement" positioning, with strong Nike representation. This segment reflects low price sensitivity and prioritizes craftsmanship, material quality, and brand prestige. It functions as a margin-driving segment and reinforces premium brand perception within the portfolio.

Segment 2: Accessible Sport Lifestyle Sun

Priced around $176, this cluster includes male-oriented sunwear with bio-injected G820 materials and a moderate base curve (~4.1) . It is largely associated with Lacoste and the "Stripes & Piping" family. These products blend lifestyle aesthetics with subtle sport cues and represent an accessible entry into branded sunwear. This segment is volume-supporting and attracts younger or moderately price-sensitive consumers seeking comfort and recognizable brand styling.

Segment 3: Fashion Acetate Sunwear

With US retail pricing near $213 and a flatter base curve (~2.6), this segment reflects fashion-forward sunglasses made primarily from acetate . It skews female and aligns with Calvin Klein's Avantgarde styling. Consumers in this segment prioritize aesthetics and trend expression over technical performance. This cluster supports seasonal assortment refresh and fashion-driven purchasing behavior.

Segment 4: Premium Fashion Optical (Female)

This cluster includes optical frames priced around $232, primarily acetate-based, and positioned within Avantgarde styling families . The standard base curve (4) supports prescription wear, while the female skew emphasizes fashion-forward daily eyewear. This segment balances function and fashion, representing stable optical demand with repeat purchase potential.

Segment 5: Value Performance Sunwear

This is the lowest-priced segment (~$138 US retail) and includes bio-injected G850 materials with higher base curves (~6.0) . The geometry indicates performance influence, even when labeled under lifestyle families. Consumers in this cluster are price-sensitive but performance-oriented, prioritizing coverage, protection, and sport functionality. This segment drives entry-level participation and unit sales volume.

Segment 6: Premium Injected Sport Optical

At approximately $253 US retail, this optical segment combines bio-injected G820 materials with silicone elements . It skews male and emphasizes ergonomic comfort and durability. Unlike fashion optical segments, this group values wearability and long-term comfort. It supports premium upselling based on performance attributes rather than aesthetics.

Segment 7: Core Lifestyle Essentials Optical

Mid-tier optical frames (~$195) composed primarily of bio-injected G850 materials define this segment . Styling is versatile and practical, positioned under Lifestyle Essentials.
This segment represents reliable, everyday optical demand and forms a broad revenue base. It balances affordability and functionality, appealing to practical consumers.

Segment 8: Elite Performance Shield Sun

This is the most specialized segment, priced around $233, featuring shield shapes, the highest base curve (~6.6), titanium construction, and no RX capability. Products in this cluster are clearly performance-first, designed for athletic or outdoor activity. This segment enhances brand credibility and anchors the high-performance assortment.

Strategic Importance of Segmentation

The segmentation analysis demonstrates that frame demand is not driven solely by brand, but by combinations of price tier, geometry, material, and styling identity. Eight distinct product-consumer archetypes emerge across the assortment, spanning premium lifestyle optical, fashion-driven sunwear, core essentials, and high-performance sport categories.

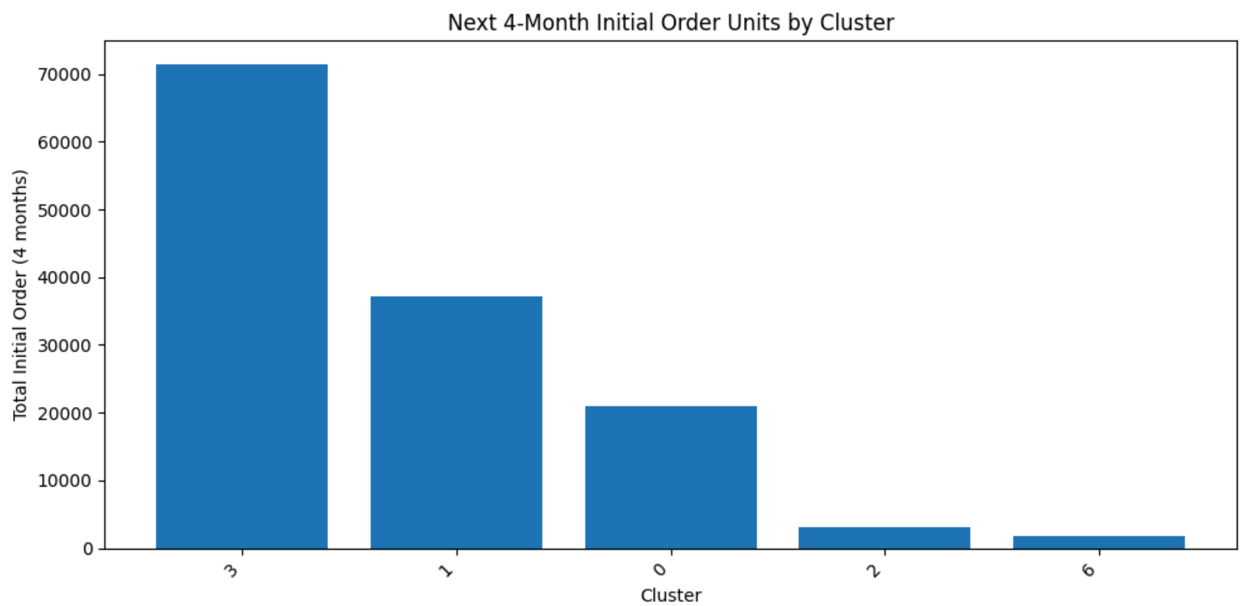
This segmentation is strategically important for several reasons:

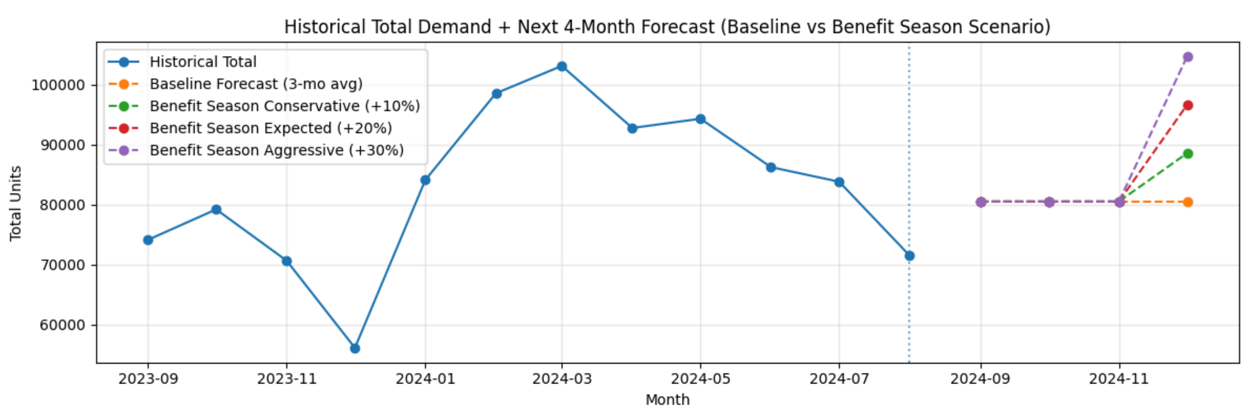
1. **Forecasting Improvement**: Rather than forecasting demand at the individual SKU level (which is highly volatile), frames can be forecasted within segment-level demand structures.
2. **Assortment Balance**: Inventory can be allocated proportionally across premium, value, fashion, and performance segments to ensure representation for each consumer type.

3. **Margin Optimization**: High-price lifestyle optical segments drive profitability, while value-performance sun segments drive volume.
4. **Regional Adaptation**: Segments can be analyzed geographically to understand which markets over-index in fashion versus performance.
5. **New Product Launch Strategy**: New frames can be positioned intentionally within high-performing segments rather than introduced without structural context.

Overall, the clustering analysis reveals that consumer preferences in eyewear are structured around clear archetypes defined by price sensitivity, aesthetic orientation, performance needs, and material expectations. These segments provide a foundation for demand forecasting, assortment planning, and strategic product positioning.

To provide full transparency on the dataset's integrity, this report begins with a breakdown of data sources and a detailed justification for the logic used to bridge gaps in the raw style array.





Data & Modeling Approach

Our Methodology

We built a machine learning model spanning 12 months of sales data across 3 brands and 162 SKUs, transforming transactional records into forward-looking demand signals.

12 Months of Data	162 Total SKUs	3 Brands
Sep 2023 through Aug 2024	Across optical and sunglass categories	Nike · Lacoste · Calvin Klein

How We Prepared the Data for Modeling

Data preparation followed four sequential steps before any model was trained:

1. **Cleaning & Standardizing:** Fixed size typos (3.0 → 53mm, 5.0 → 55mm), standardized text casing across all categorical columns, and filled missing color and material fields by merging a second attributes table on StyleCode.
2. **Reshaping to Long Format:** Pivoted 12 monthly sales columns into one row per SKU–Region–Month, giving the model a time-series structure where each row represents a single observation at a single point in time.
3. **Feature Engineering:** Built 33 features using only past data to prevent leakage: lag features (1, 2, 3, 6 months prior), rolling averages (3- and 6-month), expanding SKU statistics (max, cumulative, mean), month-over-month growth rate, brand-level context, and SKU-vs-brand ratio.
4. **Log-Transform Target:** Sales was heavily right-skewed - a few SKUs sold over 1,000 units per month while most sold around 50. Applying log(Sales+1) compressed outliers so the model could learn demand patterns across the full range of SKUs rather than overfitting to top performers.
5. **ColorDescription** was dropped (too many unique values, no predictive signal) and StyleCode was excluded to prevent data leakage - retaining it would allow the model to memorize SKU identities rather than learn generalizable demand patterns.

We Compared 3 Models - XGBoost Won

Three model types were trained and evaluated on the same log-transformed target and the same held-out test period.

Model	R^2 (Log Scale)	Result
Linear Regression	0.310	Baseline floor
Random Forest	0.520	Strong benchmark

XGBoost (Baseline)	0.580	Good out-of-the-box
XGBoost (Tuned) — Selected	**0.737**	**★ Best model**

XGBoost was selected as the final model for four reasons: it handles non-linear relationships naturally (sales does not move in straight lines); it supports mixed data types after label encoding; its built-in regularization prevents overfitting on a 162-SKU dataset; and gradient boosting means each tree specifically corrects the errors of the previous one, making it more precise on structured tabular data with temporal patterns.

R^2 (Log Scale)	R^2 (Raw Units)	MAE (Units/Month)	Input Features
0.737	**0.542**	**26.4**	**33**
Primary metric · explains 65% of demand variance	Harder benchmark due to high-volume outliers	Avg prediction off by ~29 units per SKU/month	Lag, rolling avg, brand, region, category & more

Rigorous Train / Test Split

The data was split on a strict time boundary: training on September 2023 through May 2024, and testing on June through August 2024. This replicates real-world forecasting conditions where a model trained on historical data must predict genuinely unseen future months.

- **No data leakage:** future months were never seen during training. The model had to genuinely predict the final 3 months.
- **3-fold cross-validation:** tuning scores were validated across multiple data cuts, not a single split, to ensure parameter stability.
- **Dual R^2 reporting:** log-scale R^2 (0.737) for model fit, raw-unit R^2 (0.542) for business accuracy. Both are reported honestly.
- **MAE of 29 units/month:** for SKUs averaging 50–300 units per month, this is an actionable margin for purchasing decisions.

Key Insights derived from our Models

General Insights

Rank	Feature	What It Means
#1	**Rolling 6-Month Avg Sales**	Sales trajectory is the dominant signal
#2	SKU Historical Max Sales	Peak capacity reference for each SKU

#3	SKU Expanding Average Sales	Full historical average since launch
#4	Brand (Nike / Lacoste / Calvin Klein)	Brand equity drives structural demand
#5	Optical vs Sunglass	Category type sets the demand ceiling

Key insight: Past sales patterns explain demand better than any product attribute. Brand matters more than shape, material, or size.

Frame Shapes

Frame shape demand is highly concentrated. Rectangle and Square together account for 82% of predicted volume, reflecting a consistent patient preference for classic, professional frames. This concentration is stable and should anchor the assortment. However, three niche shape signals stand out.

Frame Shape	Demand Share	Implication
Rectangle	43.8%	Core anchor - maintain depth and size coverage
Square	36.3%	Second anchor - Lacoste's top shape globally
Round	16.8%	Growing - Nike's female Round SKUs are top performers
Oval	1.3%	1 SKU, 9,400 units - most obvious white space in the assortment
Cat Eye	0.5%	2 SKUs only - trending nationally, Visionworks is behind
Aviator	1.4%	Niche but stable - primarily Lacoste Suns

Pricing

Price tier analysis reveals a clear sweet spot in the $200–$249 range, which captures the highest share of demand while also delivering strong per-SKU efficiency. The $150–$199 tier is significantly over-assorted relative to its revenue contribution.

Price Band	Demand Share	SKU Count	Revenue / SKU	Assessment
$100–$149	24.7%	24	$8,008	Efficient
$150–$199	29.4%	64	$4,017	Over-assorted

$200–$249	**32.6%**	**48**	**$6,326**	**Sweet Spot**
$250–$299	12.2%	24	$5,992	Strong
$300+	1.1%	2	$7,068	Untapped

Geography

Geographic analysis confirms that AMER and EMEA are structurally distinct markets with different brand hierarchies, size preferences, and price behaviours. Buying them with the same assortment leaves significant demand on the table in both regions.

Dimension	AMER · 56% of demand · 133,765 units	EMEA · 44% of demand · 104,434 units
#1 Brand	**Nike (61% of AMER)**	**Lacoste (39% of EMEA)**
#2 Brand	Calvin Klein (19%)	Nike (33%)
#3 Brand	Lacoste (16%)	Lacoste Sun (14%)
Top Size	54mm - broader faces	53mm - smaller, more fitted
Price Mix	Heavy in $150–$199 tier	Stronger $200–$249 pull
Sunglass	Low - prescription-first market	Higher - fashion-driven

Demand Forecasting

238,199 Units Predicted - Here Is Where They Come From

The model projects 238,199 total units across the next 4 months. Nike Optical leads forecast volume by a significant margin, though Lacoste Optical delivers the most efficient output per SKU.

Brand	Share	Forecast Units (4mo)	Avg / Month / SKU
Nike Optical	51.3%	122,266	72.6
Lacoste Optical	23.3%	55,411	54.8
Calvin Klein Optical	15.2%	36,156	40.9
Lacoste Sun	7.4%	17,609	37.0
Calvin Klein Sun	2.1%	5,012	11.8
Nike Sun	0.7%	1,745	36.4

TOTAL	100%	238,199	—

Top 10 SKUs to Prioritise — by Predicted 4-Month Volume

The following SKUs are ranked by model-predicted demand over the next 4 months. Hero-tier SKUs should be prioritised for full initial order quantities; High-tier SKUs warrant strong replenishment planning.

#	SKU	Brand	Region	Size	Colour	Pred/Mo	4Mo Qty	Priority
1	NIKE 7280	Nike	AMER	50mm	Gunsmoke	324	1,294	**HERO**
2	NIKE 7280	Nike	AMER	50mm	Clear	314	1,255	**HERO**
3	L2707	Lacoste	EMEA	53mm	Blue Steel	297	1,186	**HERO**
4	NIKE 7240	Nike	AMER	55mm	Grey	262	1,047	**HERO**
5	NIKE 7287	Nike	AMER	54mm	Matte D	251	1,003	**HERO**
6	NIKE 8054	Nike	AMER	55mm	Satin G	229	915	**HIGH**
7	NIKE 7160	Nike	AMER	55mm	Black/C	218	871	**HIGH**
8	NIKE 7160	Nike	AMER	55mm	Crystal	215	860	**HIGH**
9	NIKE 7240	Nike	AMER	53mm	Grey	201	803	**HIGH**
10	NIKE 4315	Nike	AMER	55mm	Satin N	198	794	**HIGH**

Action Items Based on Data Analysis & Demand Predictions

1. Rationalize the $150–$199 Tier

The $150–$199 band contains 64 SKUs but generates only $4,017 per SKU, compared to $6,326 per SKU in the $200–$249 tier - making it 57% less efficient.

Action:
Remove 10–15 low-velocity SKUs in this tier, particularly:

- Nike AF-series Metal Rectangles
- Overlapping Calvin Klein mid-tier styles

This will reduce cannibalization, free board space, and increase overall assortment productivity.

2. Reallocate Open-to-Buy to the $200–$249 Sweet Spot

The $200–$249 tier:

- Captures 32.6% of demand
- Has strong per-SKU efficiency
- Outperforms all other tiers structurally

Action:
Shift capital freed from mid-tier rationalization into expanding high-performing styles in the $200–$249 range. This tier balances patient willingness-to-pay and premium perception without excessive SKU dilution.

3. Expand Lacoste in AMER at Core Sizes

Lacoste generates $10,005 per SKU in the $200–$249 tier, the best-performing brand-tier combination in the portfolio.

However:

- Lacoste represents only 16% of AMER demand
- Lacoste represents 39% of EMEA demand

Action:
Add 4-6 Lacoste Square and Rectangle styles in 54–55mm for AMER. This addresses a clear regional underrepresentation gap while leveraging a proven high-efficiency brand.

4. Protect and Floor Hero SKUs Ahead of Insurance Peak

Demand spikes 73-80% from December to March due to insurance resets. The top 5 forecasted SKUs predict 1,003-1,294 units per month, representing a disproportionate share of expected volume.

Action:
Ensure uninterrupted inventory and full floor placement of all Hero-tier SKUs prior to the January–March peak. Stockouts during peak months represent direct revenue loss.

5. Test Oval White Space

Only 1 Oval SKU exists, yet it generated **9,432 units**, the highest average per SKU of any shape.

Action:
Introduce 2–3 Oval optical styles in both AMER and EMEA. This represents the clearest white-space opportunity in the assortment.

What could have made your model more accurate?

The XGBoost (Extreme Gradient Boosting) model was trained using the 9/2023–8/2024 historical demand sample combined with frame-level product attributes from the ATP files. This approach is strong for tabular retail demand problems because it can learn non-linear relationships and interactions (for example, price tier × region × sun versus optical) that are common in eyewear sales. However, the model's accuracy is ultimately bounded by the completeness of the features available at forecast time. The most impactful improvements would come from adding the drivers that sit outside basic product metadata and better aligning evaluation to how new-season forecasts are actually used.

1. **Add customer/segment signals to model "who buys what," not just "what is sold."**

Right now, the model primarily uses product attributes (brand, size, color, materials, sun/optical) and a coarse geography field. In practice, eyewear demand is shaped heavily by buyer segments—age band, plan type, first-time vs repeat purchasers, and channel (online vs store). Without these signals, the model learns an "average shopper" and can misestimate frames that appeal strongly to a specific group. Adding segment-level features would let the model learn segment preference patterns such as color families, size curves, and sun versus optical mix, and it would allow VSP to tailor initial order depth to the expected buyer mix rather than using one blended assumption across all frames.

2. **Increase geographic granularity beyond AMER/EMEA to reduce inventory misallocation.**

Geography emerged as an important differentiator in our modeling, which makes intuitive sense: climate, fashion preference, assortment strategy, and benefits timing vary by market. AMER/EMEA is too broad to capture these effects, so the model is forced to treat very different markets as similar. Replacing AMER/EMEA with country/market clusters (or distribution region groupings) would allow the model to learn local demand baselines and reduce errors caused by over-ordering in low-demand areas and under-ordering in high-demand areas. From a supply chain perspective, this directly reduces transfers, stockouts, and rushed replenishment.

3. **Enrich product "metadata" with commercial and launch context to explain demand spikes and weak launches.**

Product attributes are necessary but not sufficient for new frame forecasting. Demand for a new frame is also driven by how it is launched: campaign exposure, featured placement, email/push timing, channel placement (online-only vs widely distributed), and pricing actions (discounts, tier changes). When these factors are missing, the model may incorrectly attribute demand to product design alone and struggle to forecast items whose performance was "helped" or "hurt" by go-to-market decisions. Adding launch/marketing flags and commercial context variables would improve accuracy and also make the forecast more actionable—VSP can distinguish between inherently strong items and items that need marketing support to hit sales targets.

4. **Integrate benefit renewal timing as an explicit feature set**

Eyewear is a benefits-driven category; customers often accelerate purchases ahead of benefit resets or open enrollment windows. If the model does not explicitly encode these timing effects, it can treat predictable surges as noise and under-forecast during high-impact windows. Adding a benefit deadline calendar and features such as benefit reset month, open enrollment flags, and “days-to-expiry” indicators would help the model anticipate demand pull-forward in the 30–60 days before deadlines. Operationally, this improves initial order planning and helps VSP pre-position inventory for likely winners during demand surges, reducing stockouts and missed sales.

5. **Use an evaluation design that mirrors “new season” reality to prevent overconfidence and improve generalization.**

Because the business goal is forecasting new frames, model evaluation should reflect that deployment scenario. A random row split can unintentionally reward learning SKU-specific patterns rather than transferable drivers. A stronger approach is to hold out entire styles (StyleCode-level holdout) or use strict time-based validation that simulates forecasting forward. This produces a more realistic estimate of performance on truly new frames, highlights where the model generalizes well (for example, price tier and sun/optical effects), and clarifies limitations. In turn, it leads to better decisions about safety stock, scenario planning, and which items require closer monitoring after launch.

Contribution of Work:

1. Aneri - Data cleaning, populating missing values and preparing it for modelling. Helped with formulating recommendations. Introduction, Market Analysis, Stakeholder Analysis, Consumer journey for the presentation.
2. Tamara - Exploratory Data Analysis and Customer Segmentation Analysis (Code, Presentation, Final report)
3. Litesha - Built the Random Forest demand forecasting model for benchmarking, generated cluster-level demand forecasts and future-month projections (using the team's XGBoost outputs), and recommendations for inventory strategy.
4. Sumreen - Developed the demand forecasting modeling using XGBoost, conducted advanced data analysis to extract key insights, generated future demand predictions, and developed the final action plan and strategic recommendations based on model outputs.

APPENDIX

1. All files mentioned in the document are present in this folder, including the dataset files and the python notebook files:
   https://drive.google.com/drive/folders/1uqd3uldEmBejpZtlXsxiKsxpmFlpnhR3?usp=sharing